

Replicated Swarm Manager Action

Swarm Action

Server Dist Along Spline

float Predicted Dist Along Spline

Speed

Spline Comp

void Move Along Spline(L)

Predicted Dist = Speed $\times$ Time - last Server update time.

Predicted Dist = Server Dist Along Spline + Predicted dist

Swarm Manager Action



level placed = spawns on both
the connection is
made -

On Begin Play



if (Has Auth) set StartLoc, EndLoc

Construct Spline only on Server



Start/End Replicate to Client

On_Rep_UpdateLoc

Construct Spline On Client

Swarm Manager

Begin Play()

Construct Spline()

Spawn Swarm()

LocSet()

onServer

Replicated without
SplineToFollow

OnRep_LocSet
Construct Spline()

Spline to follow
that was set was
on server, or client
is null ptr.

To resolve OnBeginPlay
if (!SplineToFollow) chose with
replicated SplineIndex.

To DO:

On Begin Play

TArray< float > Noise

float SampleNoiseAt(1DPos)

AtEach Tick

Cache & Reverse GetWorld. GetGameState. GetServer Time Seconds

At SWARM Actor's Tick

while Predict Location

add additional 3d noise to the Final Predicted location

ASC & light height Attrib Set is given & united at a distance

